

Facility HVAC — IoT-Link

Commissioning & Design Review

2 air handlers · 2 condensers · 2 Optas · EZ Meter PQ

MooreVIEW project: facility-hvac-iot-link
ST program: st/logic/facility_hvac_iot_link.st
Firmware: MooreviewOptaMqttSt 2.3.81
Project tags: 131 · drivers: 3 · saved 2026-08-14
Generated: 2026-08-14

DRAFT FOR REVIEW — Private LAN / Starlink site

Site architecture

Phase 1: WiFi Opta Parc devices on wired LAN behind Starlink. MQTT to IoT-Link hub only — no inbound ports required. Air handlers and condensers are grouped independently (not matched AHU+cond pairs).

Pocket	Device ID	Hardware	MooreVIEW driver
Dual AHU	hvac_ahu_01	Opta + D1608E	opta_ahu (remote ST)
Dual condenser	hvac_cond_01	Opta + D1608E	opta_cond (/mcsa)
Main meter	dds_rgb	EZ Meter DDS-RGB RS-485	dds_rgb Modbus RTU
Hub	IoT-Link	MooreVIEW appliance	MQTT Parc + runtime

Network & MQTT

- Opta MQTT broker = IoT-Link LAN IP (never 127.0.0.1 on field devices).
- Topic prefix: mooreview/v1 · global site key: 1.
- Import data/projects/facility-hvac-iot-link.est.zip on IoT-Link; enable MQTT Parc hub; start runtime.

EZ Meter — facility PQ

Setting	Value
Serial port (IoT-Link PORT A)	/dev/ttyLP6
Modbus RTU	19200 8N1, slave ID 1
Service	Single-phase 240 V @ 400 A
PQ nominal / UV / OV	240 V / 216 V / 264 V
Rollups	MECH_PQ_ * tags + FACILITY_ALM on MECH_PQ_ALM

AHU Opta — hvac_ahu_01

Base Opta I1–I4 blower CT (/ct-cal). Base I5–I8 supply/return NTC. D1608E X1_I1–I2 leak digital inputs (/ahu-env).

Blower CT — calibrate on /ct-cal

Input	Function	MooreVIEW tag	Cal page
I1	Unit 1 fan start CT	AHU1_START_AMPS	/ct-cal
I2	Unit 1 fan run CT	AHU1_FAN_AMPS	/ct-cal
I3	Unit 2 fan start CT	AHU2_START_AMPS	/ct-cal
I4	Unit 2 fan run CT	AHU2_FAN_AMPS	/ct-cal

NTC — calibrate on /ahu-env (point selector)

Point ID	Input	Label	Default Vsupply	Rfixed	R25	Beta	Tag
ntc0	I5	AHU1 supply	24 V	10000 :•	10000 :•	3950	AHU1_SUPPLY_TEMP_F
ntc1	I6	AHU1 return	24 V	10000 :•	10000 :•	3950	AHU1_RETURN_TEMP_F
ntc2	I7	AHU2 supply	24 V	10000 :•	10000 :•	3950	AHU2_SUPPLY_TEMP_F
ntc3	I8	AHU2 return	24 V	10000 :•	10000 :•	3950	AHU2_RETURN_TEMP_F

Leak detection — D1608E digital DI (/ahu-env)

Point ID	Input	Label	Type	Tag
leak0	X1_I1	AHU1 pan leak	digital DI	AHU1_PAN_LEAK
leak1	X1_I2	AHU2 pan leak	digital DI	AHU2_PAN_LEAK

AHU env calibration workflow (/ahu-env)

1. Flash MooreviewOptaMqttSt; set device ID hvac_ahu_01; broker = IoT-Link LAN IP.
2. /setup !' scan D1608E in expansion slot 1.
3. Open /ahu-env !' enable AHU env processor (NV).
4. Select each NTC point (ntc0–ntc3); verify live °F; adjust Vsupply, Rfixed, R25, Beta, offset; Save.
5. Select each leak point (leak0–leak1); verify digital DI; Save.
6. Open /ct-cal !' zero/span I1–I4 blower run and start CT.
7. Deploy logic/facility_hvac_iot_link.st via IoT-Link remote execution on opta_ahu.

Condenser Opta — hvac_cond_01 (dual unit)

One Opta + D1608E monitors both condenser units. Base I1–I8: fan and compressor run/start CT (1P+cap ×4 motors). Refrigerant NTC on D1608E X1_IRAW1–4 !' T1_C–T4_C.

Motor	Run CT	Start CT	MCSA asset
U1 fan	I2	I1	fan1
U1 comp	I4	I3	fan2
U2 fan	I6	I5	comp1
U2 comp	I8	I7	comp2

D1608E	Function	Device tag	Project tag
X1_IRAW1	Unit 1 low side	T2_C	COND1_LO_TEMP_F
X1_IRAW2	Unit 1 high side	T1_C	COND1_HI_TEMP_F
X1_IRAW3	Unit 2 low side	T4_C	COND2_LO_TEMP_F
X1_IRAW4	Unit 2 high side	T3_C	COND2_HI_TEMP_F

MCSA setup — Preset: facility condenser

1. Flash Opta, set device ID hvac_cond_01, broker = IoT-Link LAN IP.
2. /setup !' scan D1608E slot 1.
3. /mcsa !' Preset: facility condenser (4 motors, 4 NTC on D1608E).
4. Verify COND1/COND2 fan+comp amps and T1_C–T4_C in telemetry.

Structured text — facility_hvac_iot_link.st

- AHU1/2: CT overload (H_CT_AMPS), leak (AHU*_PAN_LEAK), supply high temp (H_SUPPLY_HI_F) !' AHU*_FAIL_ALM.
- Condenser: MCSA faults + refrigerant temp trips !' COND*_FAIL_ALM.
- SITE_ALM = any HVAC fail; FACILITY_ALM = SITE_ALM or MECH_PQ_ALM.
- EZ Meter PQ: undervolt, overvolt, phase loss, imbalance, low PF, frequency !' MECH_PQ_ALM.

Alarm rollup

Alarm	Sources
AHU1_FAIL_ALM	Pan leak, blower run/start CT fault, supply temp high
AHU2_FAIL_ALM	Same pattern for AHU 2
COND1_FAIL_ALM	MCSA fan/comp fault, refrigerant temp trip
COND2_FAIL_ALM	MCSA faults, temp trip
SITE_ALM	Any AHU or condenser fail alarm
MECH_PQ_ALM	EZ Meter PQ thresholds
FACILITY_ALM	SITE_ALM or MECH_PQ_ALM

Default thresholds (commissioning constants)

Tag	Default	Purpose
H_CT_AMPS	18 A	Blower CT overload
H_SUPPLY_HI_F	105 °F	Supply air high temp
H_COND_HI_F	350 °F	Condenser high-side trip
H_COND_LO_F	30 °F	Condenser low-side trip
MECH_PQ_CFG_NOM_V	240 V	PQ nominal
MECH_PQ_CFG_UV_V	216 V	Undervoltage (90%)
MECH_PQ_CFG_OV_V	264 V	Overvoltage (110%)

End-to-end commissioning checklist

1. Build project: node scripts/facility-hvac-iot-link/generate-est.js
2. Import facility-hvac-iot-link.est.zip on IoT-Link
3. Wire EZ Meter RS-485 to PORT A; confirm Modbus reads at addr 1 / 19200
4. Commission hvac_ahu_01: D1608E, /ct-cal, /ahu-env
5. Commission hvac_cond_01: D1608E, /mcsa facility condenser preset (dual unit)
6. Start runtime; verify all tags GOOD; confirm SITE_ALM and FACILITY_ALM clear at rest
7. Document calibrated NTC/leak values from /ahu-env API for site records

Project artifacts

- JSON: data/projects/facility-hvac-iot-link.est.json

- Import zip: data/projects/facility-hvac-iot-link.est.zip
- Fixtures: st/fixtures/tags.facility_hvac_iot_link.json
- Starter: st/fixtures/facility_hvac_iot_link_starter.json
- Generator: node scripts/facility-hvac-iot-link/generate-est.js
- Tests: node --test test/facilityHvacIotLink.test.js
- This PDF: node scripts/facility-hvac-iot-link/build-review-pdf.js

Revision

Document generated 2026-08-14. Firmware baseline MooreviewOptaMqttSt 2.3.81. Reference: scripts/facility-hvac-iot-link/README.md · docs/facilities/EZMETER_FACILITY_PQ.md